

Apply filters to SQL queries

Project description

This project focuses on filtering out data while using SQL databases e.g. SELECT, AND, FROM and BETWEEN to improve efficiency by only showing data that you currently need and in situations where time is a factor, using filters can also reduce load times and display only the information needed.

Retrieve after hours failed login attempts

The project mentions that there has been suspicious log in attempts after hours and requires you to make a SQL query to gather information about the suspicious log in attempts.

In the lab it mentions log in attempts after 18:00. To find this information as the course and lab has shown I will use the following query:

```
SELECT *  
FROM employees  
WHERE login_attempts >= '18:00';
```

This returns all log in attempts made after the time "18:00", here is an example of the provided output.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_time > '18:00' AND success = 0;  
+-----+-----+-----+-----+-----+-----+-----+  
| event_id | username | login_date | login_time | country | ip_address | success |  
+-----+-----+-----+-----+-----+-----+-----+  
| 2 | apatel | 2022-05-10 | 20:27:27 | CAN | 192.168.205.12 | 0 |  
| 18 | pwashing | 2022-05-11 | 19:28:50 | US | 192.168.66.142 | 0 |  
| 20 | tshah | 2022-05-12 | 18:56:36 | MEXICO | 192.168.109.50 | 0 |  
| 28 | aestrada | 2022-05-09 | 19:28:12 | MEXICO | 192.168.27.57 | 0 |  
| 34 | drosas | 2022-05-11 | 21:02:04 | US | 192.168.45.93 | 0 |  
| 42 | cgriffin | 2022-05-09 | 23:04:05 | US | 192.168.4.157 | 0 |  
| 52 | cjackson | 2022-05-10 | 22:07:07 | CAN | 192.168.58.57 | 0 |  
| 69 | wjaffrey | 2022-05-11 | 19:55:15 | USA | 192.168.100.17 | 0 |  
| 82 | abernard | 2022-05-12 | 23:38:46 | MEX | 192.168.234.49 | 0 |  
| 87 | apatel | 2022-05-08 | 22:38:31 | CANADA | 192.168.132.153 | 0 |  
| 96 | ivelasco | 2022-05-09 | 22:36:36 | CAN | 192.168.84.194 | 0 |  
| 104 | asundara | 2022-05-11 | 18:38:07 | US | 192.168.96.200 | 0 |  
| 107 | bisles | 2022-05-12 | 20:25:57 | USA | 192.168.116.187 | 0 |  
| 111 | aestrada | 2022-05-10 | 22:00:26 | MEXICO | 192.168.76.27 | 0 |  
| 127 | abellmas | 2022-05-09 | 21:20:51 | CANADA | 192.168.70.122 | 0 |  
| 131 | bisles | 2022-05-09 | 20:03:55 | US | 192.168.113.171 | 0 |  
| 155 | cgriffin | 2022-05-12 | 22:18:42 | USA | 192.168.236.176 | 0 |  
| 160 | jclark | 2022-05-10 | 20:49:00 | CANADA | 192.168.214.49 | 0 |  
| 199 | yappiah | 2022-05-11 | 19:34:48 | MEXICO | 192.168.44.232 | 0 |  
+-----+-----+-----+-----+-----+-----+-----+  
19 rows in set (0.021 sec)
```

Retrieve login attempts on specific dates

The project asks to follow up on login attempts next, we will need to make a new query and apply filters to see login attempts from the dates for 2022-05-09 or 2022-05-08.

Since I am expected to find two separate dates in the same table of contents I will need to use the OR operator so that it will return 2022-05-09 or 2022-05-08.

The following query needed for this is:

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-08' OR login_date = '2022-05-09';
```

This returns all login attempts made on both 2022-05-08 and 2022-05-09, here is a snippet of the provided output.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
24	arusso	2022-05-09	06:49:39	MEXICO	192.168.171.192	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
28	astrada	2022-05-09	19:28:12	MEXICO	192.168.27.57	0
30	yappiah	2022-05-09	03:22:22	MEX	192.168.124.48	1
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
39	yappiah	2022-05-09	07:56:40	MEXICO	192.168.57.115	1
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0
47	dkot	2022-05-08	05:06:45	US	192.168.233.24	1
49	asundara	2022-05-08	14:00:01	US	192.168.173.213	0
53	nmason	2022-05-08	11:51:38	CAN	192.168.133.188	1
56	acook	2022-05-08	04:56:30	CAN	192.168.209.130	1
58	ivelasco	2022-05-09	17:20:54	CAN	192.168.57.162	0
61	dtanaka	2022-05-09	09:45:18	USA	192.168.98.221	1
65	aalonso	2022-05-09	23:42:12	MEX	192.168.52.37	1
66	astrada	2022-05-08	21:58:32	MEX	192.168.67.223	1
67	abernard	2022-05-09	11:53:41	MEX	192.168.118.29	1
68	mrh	2022-05-08	17:16:13	US	192.168.42.248	1
70	tmitchel	2022-05-09	10:55:17	MEXICO	192.168.87.199	1
71	mcouliba	2022-05-09	06:57:42	CAN	192.168.55.169	0
72	alevitsk	2022-05-08	12:09:10	CANADA	192.168.139.176	1
79	abernard	2022-05-09	11:41:15	MEX	192.168.158.170	0

Note: The query returned 75 rows in the set but to ensure the ability to read the provided image I chose only to include a portion of the rows.

Retrieve login attempts outside of Mexico

The project now requires me to focus on login attempts made outside of Mexico. As you can see from the previously provided images under “country” you may see there is rows with both “MEX” and “MEXICO”. I will need to use the correct filters to make sure to include both.

The following query needed for this request is:

```
SELECT *  
FROM employees  
WHERE NOT country LIKE 'MEX%';
```

Note: I feel this query needs clarifying. Since this part requires you to find login attempts made outside of Mexico we need to use the NOT operator, this will show us every login attempt that did not occur from Mexico. The LIKE operator is used with the wildcard operator “%” since using these together in the query “LIKE ‘MEX%’” will exclude any rows containing both MEX and MEXICO.

```
MariaDB [organization]> SELECT *  
->  
-> FROM log_in_attempts  
->  
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0
5	jrafael	2022-05-11	03:05:59	CANADA	192.168.86.232	0
7	eraab	2022-05-11	01:45:14	CAN	192.168.170.243	1
8	bisles	2022-05-08	01:30:17	US	192.168.119.173	0
10	jrafael	2022-05-12	09:33:19	CANADA	192.168.228.221	0
11	sgilmore	2022-05-11	10:16:29	CANADA	192.168.140.81	0
12	dkot	2022-05-08	09:11:34	USA	192.168.100.158	1
13	mrhah	2022-05-11	09:29:34	USA	192.168.246.135	1
14	sbaelish	2022-05-10	10:20:18	US	192.168.16.99	1
15	lyamamot	2022-05-09	17:17:26	USA	192.168.183.51	0
16	mcouliba	2022-05-11	06:44:22	CAN	192.168.172.189	1
17	pwashing	2022-05-11	02:33:02	USA	192.168.81.89	1
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
19	jhill	2022-05-12	13:09:04	US	192.168.142.245	1
21	iuduike	2022-05-11	17:50:00	US	192.168.131.147	1
25	sbaelish	2022-05-09	07:04:02	US	192.168.33.137	1
26	apatel	2022-05-08	17:27:00	CANADA	192.168.123.105	1
29	bisles	2022-05-11	01:21:22	US	192.168.85.186	0
31	acook	2022-05-12	17:36:45	CANADA	192.168.58.232	0
32	acook	2022-05-09	02:52:02	CANADA	192.168.142.239	0
33	zbernal	2022-05-11	02:52:10	US	192.168.72.59	1
34	drosas	2022-05-11	21:02:04	US	192.168.45.93	0
36	asundara	2022-05-08	09:00:42	US	192.168.78.151	1
37	eraab	2022-05-10	06:03:41	CANADA	192.168.152.148	0
38	sbaelish	2022-05-09	14:40:01	USA	192.168.60.42	1
41	apatel	2022-05-10	17:39:42	CANADA	192.168.46.207	0
42	cgriffin	2022-05-09	23:04:05	US	192.168.4.157	0
43	mcouliba	2022-05-08	02:35:34	CANADA	192.168.16.208	0
44	daquino	2022-05-08	07:02:35	CANADA	192.168.168.144	0

Retrieve employees in Marketing

For the next task I am told it is my responsibility to find the machines belonging to the Marketing department in the East offices for security updates.

For this the query I need to make must include the department and the offices with a wildcard:

```
SELECT *  
FROM employees  
WHERE department = 'Marketing' AND office LIKE 'East%';
```

Note: As seen in the query above, I filtered for only the Marketing department and then further filtered with the AND operator to filter for offices starting with East.

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE department = 'Marketing' AND office LIKE 'East%';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
| 1000 | a320b137c219 | elarson | Marketing | East-170 |  
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |  
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |  
| 1103 | NULL | randers | Marketing | East-460 |  
| 1156 | a184b775c707 | dellery | Marketing | East-417 |  
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.227 sec)  
MariaDB [organization]>
```

Retrieve employees in Finance or Sales

Similar to the above example the project says that I need to filter again but for both the Finance Department and the Sales department, the same reason as before applies here now too, the reason is for a security update.

For this query I will need to OR operator so the query knows to get the information needed from both departments, the AND operator here would cause a syntax error because it would require both conditions to be true but each employee only belongs to one department:

```
SELECT *  
FROM employees  
WHERE department = 'Finance' OR department = 'Sales';
```

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE department = 'Finance' OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l2l2m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1029	d336e475f676	ivelasco	Finance	East-156
1035	j236k303l245	bisles	Sales	South-171
1039	n253o917p623	cjackson	Sales	East-378
1041	p929q222r778	cgriffin	Sales	North-208
1044	s429t157u159	tbarnes	Finance	West-415
1045	t567u844v434	pwashing	Finance	East-115
1046	u429v921w138	daquino	Finance	West-280
1047	v109w587x644	cward	Finance	West-373
1048	w167x592y375	tmitchel	Finance	South-288
1049	NULL	jreckley	Finance	Central-295
1050	y132z930a114	csimmons	Finance	North-468
1057	f370g535h632	mscott	Sales	South-270
1062	k367l639m697	redwards	Finance	North-180
1063	l686m140n569	lpope	Sales	East-226
1066	o678p794q957	ttyrell	Sales	Central-444
1069	NULL	jpark	Finance	East-110
1071	t244u829v723	zdutchma	Sales	West-348
1072	u905v920w694	esmith	Sales	East-421
1076	y347z204a710	fgarcia	Finance	Central-270
1078	a667b270c984	sharley	Sales	North-418

Retrieve all employees not in IT

The project informs me that the employee machines need one more update but the IT department has already received this update and needs to be excluded from the query. To do this I will simply use the NOT operator again and select departments and give the condition 'Information Technology' so the query knows to return all employees except IT.

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

```
MariaDB [organization]> SELECT *  
->  
-> FROM employees  
->  
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodriqu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1016	q793r736s288	sbaelish	Human Resources	North-229
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1020	u899v381w363	arutley	Marketing	South-351
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115
1026	a998b568c863	apatel	Human Resources	West-320
1027	b806c503d354	mrah	Marketing	West-246
1028	c603d749e374	astrada	Human Resources	West-121
1029	d336e475f676	ivelasco	Finance	East-156
1030	e391f189g913	mabadi	Marketing	West-375
1031	f419g188h578	dkot	Marketing	West-408
1034	i679j565k940	bsand	Human Resources	East-484
1035	j236k303l245	bisles	Sales	South-171
1036	k550l533m205	rjensen	Marketing	Central-239
1038	m873n636o225	btang	Human Resources	Central-260
1039	n253o917p623	cjackson	Sales	East-378
1040	o783p832q294	dtarly	Human Resources	East-237
1041	p929q222r778	cgriffin	Sales	North-208

Summary

This project shows how filters can be applied for efficient queries, and to help show only the information needed for any given task. In the above tasks you needed to filter for login times, locations and filter for departments that need updates to relevant machines. By applying operator like AND, OR, LIKE and NOT I was able to achieve this and query the correct information I needed to complete the tasks and investigate the times and dates of login times and to also check to see what machines in departments needed security updates.